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Geochemical and thermodynamical modeling of magmatic sources and processes for the Xiarihamu sulfide deposit in the eastern Kunlun Orogen, western China

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ABSTRACT

Magmatic sulfide deposits in subduction-collision settings are genetically related to oxidized and hydrous magmas, where the degree of partial melting, fractional crystallization, and sulfide saturation are all expected to differ significantly from those in dry and reduced magmas associated with most giant magmatic sulfide deposits in intracratonic settings. This study provides petrological and geochemical data as well as multiple modeling approaches for the post-collisional Xiarihamu magmatic sulfide deposit of the eastern Kunlun Orogen to investigate melting mechanism and magmatic differentiation in an oxidized and hydrous magmatic system. Petrographical observations and whole-rock geochemistry suggest that the olivine orthopyroxenite, orthopyroxenite, websterite and gabbronorite in the Xiarihamu complex were cumulates. Pressures estimated from clinopyroxene and hornblende compositions and oxygen fugacities from hornblende and biotite compositions further suggest that these cumulates crystallized under moderate- to high-pressure, oxidized conditions. The Xiarihamu parental magma was high-MgO basalt as evidenced by multiple modeling methods. This parental magma was generated from a metasomatized mantle peridotite source in the spinel stability field with moderate melting degrees and then experienced a fractional crystallization process with a first iron-enriched and then silica-enriched trend under moderate-high pressures to form the observed crystallization sequence and mineral compositions. Minor amounts of wall rock assimilation during magmatic evolution were supported by Sr-Nd isotopic modeling. Sulfide saturation previously proposed by crustal sulfur addition at Xiarihamu was not straightforward but was likely caused by an oxygen fugacity decrease triggered by magmatic differentiation.

1. Introduction

Magmatic sulfide deposits currently account for ~56% of the world's Ni production and over 96% of platinum-group element (PGE) production (Mudd and Jowitti, 2014). Most giant magmatic sulfide deposits occur in intracratonic settings associated with relatively dry and reduced magmatic systems produced by mantle plume activity (Barnes et al., 2017; Naldrett, 2010). In this system, the degree of partial melting is believed to exert a strong control on the Ni-Cu-PGE contents of mantle-derived magmas because of the different partitioning of these elements between sulfides and silicates (Arndt et al., 2005; Wang et al., 2018). Low-degree partial melting should lead to very low abundances of Pd-group PGE (PPGE) and low to moderate abundances of Ni, whereas high degrees of partial melting should produce low abundances of PPGE and high abundances of Ni and Ir-group PGE (IPGE) (Naldrett, 2010). The mantle plume-related oceanic and continental flood basaltic magmas mainly crystallize plagioclase-dominated assemblages at low-moderate pressures, which drive the evolution of magmatic liquids along the tholeiitic differentiation trend, resulting in an iron enriched trend with decreasing magnesium contents in the melt (Husen et al., 2016). The solubility of sulfides in basaltic magmas increases with decreasing pressure and thus basaltic magmas emplaced at relatively shallow depths are unlikely to be saturated in sulfides (Jégo et al., 2016; Mavrogenes and O'Neill, 1999). It has long been debated and discussed that addition of crustal sulfur by direct melting and assimilation of wall rocks and xenoliths is the most

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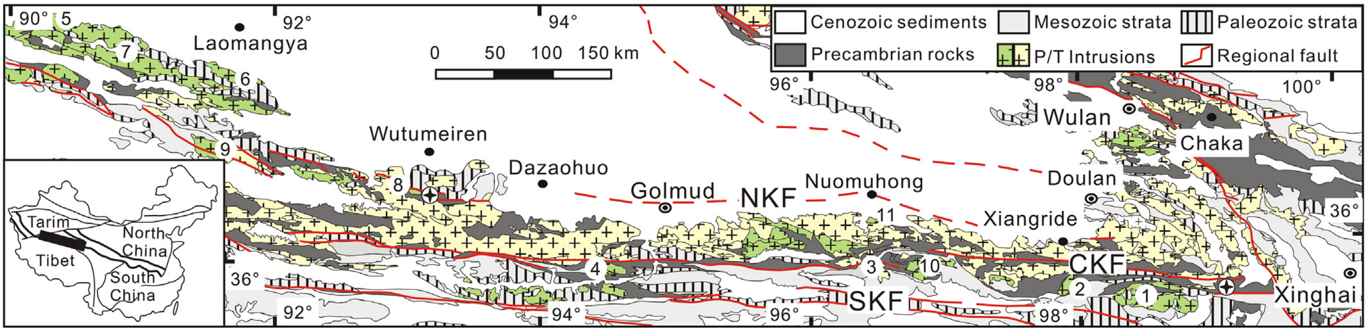

Fig. 1. Geological map of the eastern Kunlun Orogen showing the distribution of the Paleozoic (P) and Triassic (T) intrusions. Subduction-related intrusions: 1-Zhiyu granite, 447 Ma (Zhou et al., 2016), 2-Kekesha quartz diorite, 515 Ma (Zhang et al., 2010), 3-Huxiaoqin hornblende diabase, 438 Ma (Liu et al., 2013), 5-Wutoushan gneissic granite, 485 Ma, 6-Shuangshixia granite, 462 Ma and 7-Naitoushan diorite, 446 Ma (W. Li et al., 2013b). Main collisional to post-collisional intrusions: 4-Wanbaogou A-type granite, 441 Ma (Wang et al., 2012), 8-Xiarihamu A-type granite, 391 Ma (Wang et al., 2013), 9-Kayakedengtage I-type granite, 394 Ma (Chen et al., 2006), 10-Helegangnaren A-type granite, 425 Ma (R.B. Li et al., 2013a), 11-Yuejinshan S-type granite, 407 Ma (Liu et al., 2012). Stars represent high-pressure eclogites with the ages of 450–430 Ma at Wenquan in the east (Meng et al., 2015) and of 411 Ma at Xiarihamu in the west (Qi et al., 2014). NKF, the Northern Kunlun Fault, CKF, the Central Kunlun Fault, SKF, the Southern Kunlun Fault. Insert is the tectonic divisions of China showing the location of the eastern Kunlun Orogen.

efficient process to lead to sulfide saturation in this system (i.e., more efficient than fractional crystallization, addition of volatiles, magma mixing, and crustal assimilation without sulfur addition; Keays and Lightfoot, 2010; Ripley and Li, 2013, 2017; Robertson et al., 2015).

A number of magmatic sulfide deposits also occur in subduction-collision settings (Barnes et al., 2017; Naldrett, 2010), including those in the Central Asian Orogenic Belt (Gao et al., 2013; Zhao et al., 2016). They are economically subordinate but may be potentially significant with the recent discovery of the giant Xiarihamu magmatic sulfide deposit in the eastern Kunlun Orogen in western China (C.S. Li et al., 2015a; Peng et al., 2016; Song et al., 2016; Zhang et al., 2016). Parental magmas related to such sulfide deposits in subduction-collision settings are generally considered to be derived from the mantle wedge metasomatized by slab-derived fluids and melts (Gao et al., 2013; Song and Li, 2009). As such, both the parental magmas and the mantle wedge are commonly believed to be hydrous and oxidized (Richards, 2015). In this scenario less (~6%) melting in the mantle wedge may be required and high degrees of partial melting are unlikely (Wilkinson, 2013). A hydrous and oxidized arc basaltic magma should initially crystallize the mineral assemblage dominated by pyroxenes and amphibole instead of olivine and plagioclase at moderate-high pressures, which usually produces the calc-alkaline differentiation trend with the silica enrichment (Greene et al., 2006; Hamada and Fujii, 2008; Jagoutz, 2010; Pichavant and Macdonald, 2007). The arc basalts have typically higher than depleted mantle values of δ^{34}S (up to and rarely above 13‰), reflecting a contribution from marine sulfates fixed in the subducted oceanic crust and transported into the mantle wedge by metasomatic fluids (Alt et al., 1993). The δ^{34}S values for the post-collisional magmatic sulfide deposits in the Central Asian Orogenic Belt and the eastern Kunlun Orogen are similar to those of the arc basalts but are generally interpreted to reflect the addition of crustal sulfur (C.S. Li et al., 2015a; Zhang et al., 2016; Zhao et al., 2016). Considering the fact that hydrous and oxidized magmas also contain high sulfur contents (Matjuschkin et al., 2016), the necessity of crustal sulfur addition for the magmatic ore formation in subduction-collision settings is doubtful. In other words, the key steps for the formation of magmatic sulfide deposits in subduction-collision settings may be fundamentally different relative to those formed in stable intracratonic settings.

The Xiarihamu sulfide deposit in the Eastern Kunlun Orogen is the largest magmatic sulfide deposit in subduction-collision settings. Presently, a comprehensive set of published data about mineral chemistry, whole-rock major- and trace-element, PGE and Sr-Nd-Os-S isotopes as well as zircon U-Pb and Lu-Hf isotopes for the deposit indicate a controversial petrologic and ore genesis. The parental magma has been proposed to be either boninite (C.S. Li et al., 2015a) or high-MgO basalt (Du et al., 2014; Jiang et al., 2015; Song et al., 2016) originated from an enriched (Jiang et al., 2015; Zhang et al., 2016) or depleted (Wang et al., 2014) mantle by low-degree partial melting of mantle peridotite (Jiang et al., 2015; Zhang et al., 2016) or mantle pyroxenite (Song et al., 2016). Sulfide retention in the mantle source (Song et al., 2016; Zhang et al., 2016) or previous sulfide segregation during magma ascent (Du et al., 2014) has been applied to interpret the PGE depletion. Relatively uniform sulfide δ^{34}S values (2.2–6.8‰) have been exclusively explained by the addition of crustal sulfur (Du et al., 2014; C.S. Li et al., 2015a; Zhang et al., 2016). In this paper, we report on detailed field observations, mineral chemistry, whole-rock major- and trace-element compositions, and Sr-Nd isotopes of fresh mafic and ultramafic rocks from the Xiarihamu complex and simultaneously apply multiple geochemical and thermodynamical modelings to constrain the degrees of partial melting in the mantle source, the parental magma, and the feasible differentiation trend during magmatic ascent.

2. Geological setting

The Kunlun Orogen is the western segment of the Central China Orogenic Belt that separates the Tarim and North China cratons to the north from the Tibetan and the South China Block to the south (Fig. 1). It can be divided into western and eastern parts by the Altyn Fault. The eastern Kunlun Orogen, connected with the western Qinling Orogen in the east, is situated between the Qaidam Block to the north and the Hohxil-Bayanhar Block to the south. It is underlain by a Mesoarchean-Mesoproterozoic basement (Chen et al., 2011; He et al., 2016a), and has witnessed multistage subduction and collision from Neoproterozoic to Early Mesozoic (Bian et al., 2004; Chen et al., 2015; He et al., 2016b; Meng et al., 2013a; Yang et al., 1996; Zhang et al., 2012, 2014; Zhu et al., 2006).

The eastern Kunlun Orogen consists of the Northern, Middle and Southern Kunlun terranes separated by the Northern, Central and Southern Kunlun Faults (Fig. 1) (Jiang et al., 1992; Liu et al., 2005; Xu et al., 2006). Early Paleozoic high-pressure eclogites (450–411 Ma) and granulites (508 Ma) (Li et al., 2006; Meng et al., 2013b, 2015; Qi et al., 2014), and high-temperature granulites (460–430 Ma) (Yu, 2005; Zhang et al., 2003) have been reported to occur in the Precambrian metamorphic rocks of the Middle Kunlun Terrance (Fig. 1). The Central Kunlun Fault represents an Early Paleozoic suture zone containing exposed ophiolites with zircon U-Pb ages of 518–509 Ma (Feng et al., 2010; Yang et al., 1996). The Southern Kunlun Fault is considered to represent a composite suture zone and is characterized by associated ophiolites having two distinct age groups of 516–467 Ma and 345–333 Ma (Bian et al., 2004; L. Chen et al., 2001a; Liu et al., 2011).

Early Paleozoic subduction-related intrusive rocks mostly occur in the Northern Kunlun Terrane and the eastern part of the Middle and Southern Kunlun terranes (Fig. 1). They are dominated by calc-alkaline diorites, granodiorites, and monzogranites with zircon U-Pb ages of 485–422 Ma in the Northern Kunlun Terrane (W. Li et al., 2013b; Yu et al., 2017). However, calc-alkaline gabbros, diorites, quartz diorites, granodiorites, monzogranites and syenogranites with zircon U-Pb ages ranging from 515 to 436 Ma are common in the eastern part of the Middle and Southern Kunlun terranes (N.S. Chen et al., 2001b; R.B. Li et al., 2015b; Liu et al., 2013; Xiong et al., 2015; Zhang et al., 2010; Zhou et al., 2016). High-Mg diorites and granodiorites were emplaced at ~430 Ma on the south of Nuomuhong, likely indicating the beginning of collision (Zhang et al., 2014). More voluminous A-, S-, and I-type granites (441–370 Ma) (Chen et al., 2006; Chen et al., 2013; Gao et al., 2014; R.B. Li et al., 2013a; Liu et al., 2012; Lu et al., 2013; Shi et al., 2016; Wang et al., 2013; Wang et al., 2012; Yu et al., 2017) and minor amounts of diorites and gabbros (413–380 Ma) (Ao et al., 2014; Liu et al., 2012; Xiong et al., 2014; Yu et al., 2017; Zhou et al., 2016) in the eastern Kunlun Orogen suggest a collisional to post-collisional setting. The Xiarihamu ore-related mafic-ultramafic complex was emplaced at 411–406 Ma and thus was formed in a post-collisional setting (Jiang et al., 2015; C.S. Li et al., 2015a; Song et al., 2016; Wang et al., 2014).

3. Geology of the Xiarihamu magmatic sulfide deposit

The Xiarihamu sulfide deposit in the Middle Kunlun Terrane is the largest magmatic sulfide deposit in subduction-collision settings and contains ~157 million metric tons (Mt) of sulfide ores with an average grade of 0.65 wt% Ni and 0.14 wt% Cu (C.S. Li et al., 2015a; Song et al., 2016). The Xiarihamu complex of ~1.4 km wide and ~2.0 km long (including the western unexposed part) was emplaced into the gneiss, schist and marble of the Mesoarchean-Mesoproterozoic Jinshuikou Complex with gneiss xenoliths near the contacts (Fig. 2). The complex consists of an ultramafic portion and a gabbroic portion with sharp contacts. The coarse- to fine-grained ultramafic rocks principally comprise orthopyroxenite covering on olivine orthopyroxenite in the western complex and overlain by websterite in the eastern complex (Song et al., 2016). Sharp and clear contacts are not observed between the different types of ultramafic rocks (see also Jiang et al., 2015). The sulfide ores are mainly hosted in the olivine orthopyroxenite and orthopyroxenite, and consist of pyrrhotite, pentlandite, and subordinate chalcopyrite. The gabbroic rocks are medium-grained gabbronorite. Zircon U-Pb ages of 408.1 ± 2.9 Ma, 406.1 ± 2.7 Ma, and 411 ± 2.4 Ma for websterite samples and 439.1 ± 3 Ma, 431.3 ± 2.1 Ma, 423 ± 1 Ma, and 405.5 ± 2.7 Ma for gabbronorite samples have been reported (Jiang et al., 2015; C.S. Li et al., 2015a; Song et al., 2016; Wang et al., 2014). Only the gabbronorite from the northern complex that yielded a similar U-Pb age with the websterite (Song et al., 2016) is investigated in this study. The other gabbronorites in the complex with older ages are not considered in this study.

Olivine is medium-, fine-grained in size and occurs as rounded grains in contact with orthopyroxene crystals in olivine orthopyroxenite and as inclusions within orthopyroxene crystal or surrounded by several orthopyroxene crystals in orthopyroxenite (Fig. 3a). Coarse- to medium-grained orthopyroxene is commonly anhedral in olivine orthopyroxenite and orthopyroxenite (Fig. 3a), relative to its counterpart in websterite. Minor amounts of fine-grained, euhedral, and interstitial orthopyroxene in orthopyroxenite and websterite also occur (Fig. 3b). Clinopyroxene is fine-grained and interstitial in olivine orthopyroxenite and orthopyroxenite, but appears as both medium-grained, subhedral grains and fine-grained, euhedral crystals in websterite. Hornblende and biotite in olivine orthopyroxenite and orthopyroxenite, and plagioclase, hornblende and biotite in websterite are exclusively anhedral and interstitial (Fig. 3a–b). Clinopyroxene is medium-grained and severely absorbed (Fig. 3c) whereas medium- to fine-grained plagioclase is rounded or tabular in gabbronorite with minor amounts of fine-grained, prismatic clinopyroxene. Hornblende and biotite in gabbronorite are anhedral and interstitial. Some hornblende grains in gabbronorite have occasionally developed 10–15 μm wide margins with a clean and smooth surface (Fig. 3d). Petrographic observations in this and previous studies have clearly demonstrated a crystallization sequence transited from olivine + orthopyroxene to orthopyroxene and then orthopyroxene + clinopyroxene in the ultramafic rocks. Clinopyroxene + plagioclase in the gabbronorite crystallized latest.

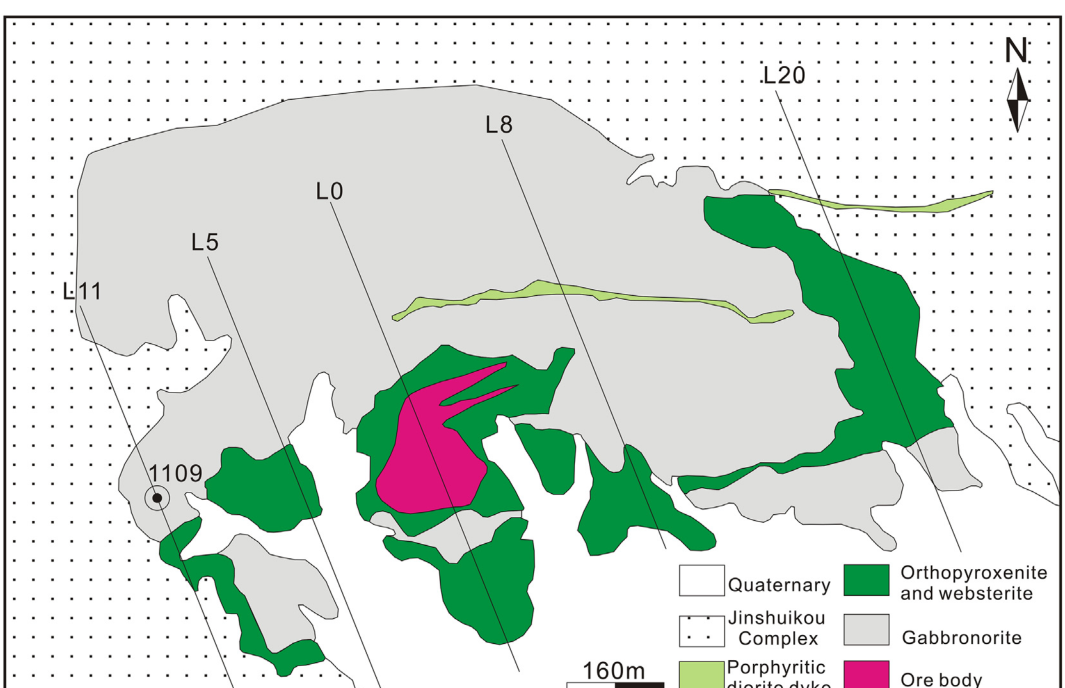

Fig. 2. Geologic map of the Xiarihamu complex modified after Song et al. (2016).

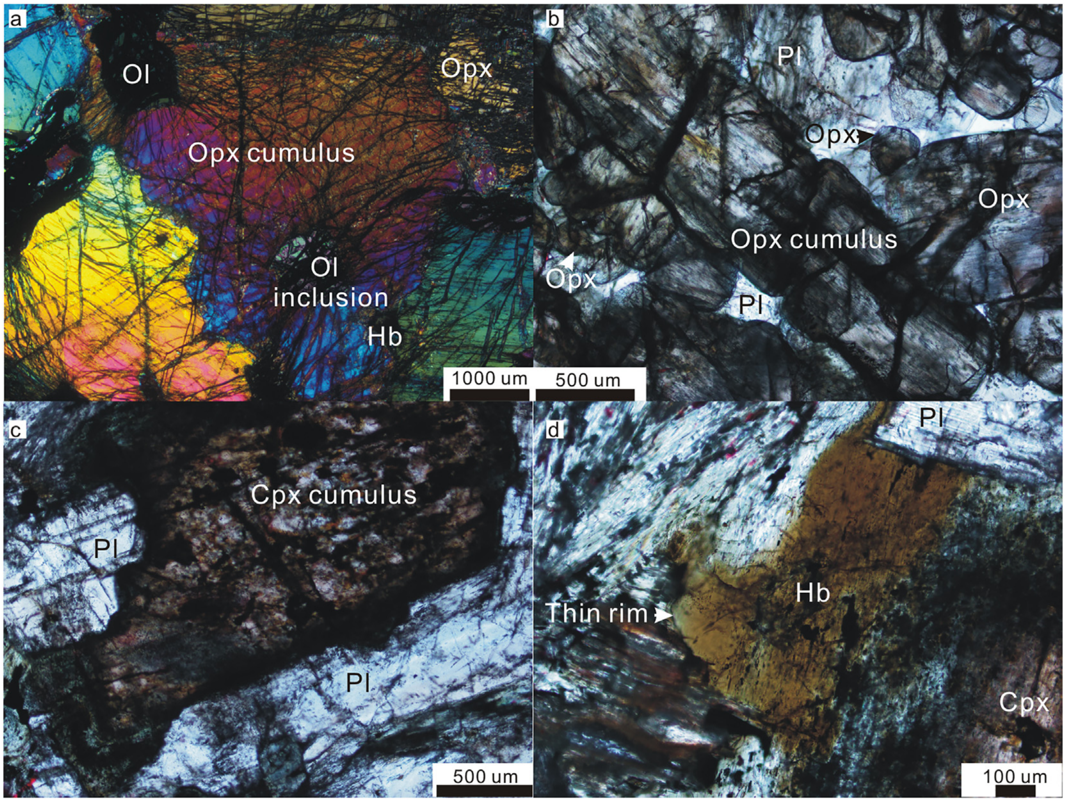

Fig. 3. Representative photomicrographs. (a) Coarse-grained and severely absorbed orthopyroxene (Opx) with olivine (Ol) inclusions filled by fine-grained and interstitial orthopyroxene and hornblende (Hb) in an olivine orthopyroxenite sample. (b) Medium-grained orthopyroxene cumulus filled by fine-grained and interstitial orthopyroxene and plagioclase (Pl) in an orthopyroxenite sample. (c) Medium-grained and severely absorbed clinopyroxene (Cpx) surrounded by plagioclase in a gabbronorite sample. (d) Interstitial hornblende having a thin margin with clean and smooth surface in a gabbronorite sample.

4. Analytical methods

4.1. Electron microprobe analyses

Three representative samples, including an olivine orthopyroxenite with coarse-grained, corroded orthopyroxene containing olivine inclusions (Fig. 3a) from drill core QY1109 (Fig. 2), and an orthopyroxenite and a gabbronorite collected from surface outcrops, were selected to mineral major oxide analysis. The orthopyroxenite sample contains both medium- to coarse-grained, slightly corroded, cumulate orthopyroxene and fine-grained and interstitial orthopyroxene (Fig. 3b). Mineral compositions were determined at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences (Wuhan), with a JEOL JXA-8100 Electron Probe Micro Analyzer equipped with four wavelength-dispersive spectrometers. Polished thin sections of the samples were firstly coated with a thin conductive carbon film prior to analysis. The precautions suggested by Zhang and Yang (2016) were used to minimize the difference of carbon film thickness between samples and a ca. 20 nm approximately uniform coating was obtained. During the analysis, an accelerating voltage of 15 kV, a beam current of 20 nA and a 10 μm spot size were used to analyze minerals. Data were corrected on-line by using a modified ZAF (atomic number, absorption, fluorescence) correction procedure. The peak counting time was 10 s for Na, Mg, Al, Si, K, Ca, Fe, P and 20 s for Ti and Mn. The background counting time was one-half of the peak counting time on the high- and low-energy background positions. The following standards were used: sanidine (K), pyrope garnet (Fe, Al), diopside (Ca), jadeite (Na), rhodonite (Mn), forsterite (Si, Mg), rutile (Ti), fluoapatite (P). Analytical errors are estimated to be < 1%. Calculations of mineral formulas were accomplished with the Minpet Geological Software (version 2.02).

4.2. Whole-rock major element analyses

Fresh whole-rock samples without visible sulfide mineralization and hydrothermal alteration from the ultramafic rocks and the gabbronorite with the age of 405.5 ± 2.7 Ma in the northern complex were collected. They were firstly crushed in a steel crusher and then powdered to 200-mesh by using an agate mill. Major element compositions were determined at the Australian Laboratory Services Chemex Co. Ltd., Guangzhou, China. Ferrous iron was analyzed by acid decomposition and titration with potassium dichromate. The samples were digested with sulfuric and hydrofluoric acids and then titrated in a beaker containing diluted sulfuric acid, orthophosphoric acid and boric acid with potassium dichromate solution. Other major elements were analyzed by X-ray fluorescence (XRF) spectrometry. Prior to major element analysis, calcined or ignited samples (~0.9 g) were added to ~9.0 g of lithium borate flux (50-50% $Li_2B_4O_7$-$LiBO_2$) and were mixed well before being fused in an autofluxer at temperatures between 1050 °C and 1100 °C. The resulting molten glass was cast into a flat glass disk that was used for X-ray fluorescence spectrometry (XRF) analysis by applying an ME-XRF26 instrument, yielding analyses with precision values better than ± 1–2%.

4.3. Whole-rock trace element analyses

Trace element analysis was conducted on Agilent 7700e ICP-MS at the Wuhan SampleSolution Analytical Technology Co., Ltd., Wuhan, China. The detailed sample-digesting procedure was as follows: (1) powder samples (200-mesh) were placed in an oven at 105 °C for drying of 12 h; (2) 50 mg sample powder was accurately weighed and placed in a Teflon bomb; (3) 1 ml HNO_3 and 1 ml HF were slowly added into the Teflon bomb; (4) the Teflon bomb was then putted in a stainless steel pressure jacket and heated to 190 °C in an oven for > 24 h; (5) after cooling, the Teflon bomb was opened and placed on a hotplate at 140 °C and evaporated to incipient dryness, and then 1 ml HNO_3 was added and evaporated to dry again; (6) 1 ml HNO_3, 1 ml MQ water and 1 ml internal standard solution of In (1 ppm) were added, and the Teflon bomb was resealed and placed in the oven at 190 °C for > 12 h; (7) the final solution was transferred to a polyethylene bottle and diluted to 100 g by the addition of 2% HNO_3 for ICP-MS analyses. Relative standard deviations estimated from randomly-selected repeated analyses and four reference materials (AGV-2, BHVO-2, BCR-2, and RGM-2) are better than 5% for rare earth elements and 5–12% for other elements.

4.4. Whole-rock Sr-Nd isotope analyses

Whole-rock Sr-Nd isotopic compositions were obtained on a Finnigan Triton thermo-ion mass spectrometer at the State Key Laboratory for Mineral Deposit Research, Nanjing University following the methods of Pu et al. (2005). The measured $^{87}Sr/^{86}Sr$ and $^{143}Nd/^{144}Nd$ ratios are normalized to $^{86}Sr/^{88}Sr = 0.11940$ and $^{146}Nd/^{144}Nd = 0.721900$, respectively. Total analytical blanks are 5×10^{-11} g for Sm and Nd and $(2–5) \times 10^{-10}$ g for Rb and Sr. The average $^{143}Nd/^{144}Nd$ ratio of the JNDi-1 standard measured during the sample runs was 0.512095 ± 5 (2σ), whereas the NBS-987 standard gave $^{87}Sr/^{86}Sr = 0.710253 \pm 5$ (2σ).

5. Results

5.1. Mineral chemistry

Previous studies provided a large quantity of mineral analyses for the Xiarihamu complex, but mineral compositional variations between different rock types and between different mineral phases were not emphasized (Jiang et al., 2015; C.S. Li et al., 2015a). Orthopyroxene obviously varies in the enstatite (En) content among different rock types from $En_{81–88}$ in olivine orthopyroxenite and orthopyroxenite to $En_{79–84}$ in websterite (Fig. 4a). However, the compositions from orthopyroxene of different textural types in the same rock types show limited variations, such as coarse-grained orthopyroxene cumulus being compositionally similar with its fine-grained interstitial counterpart in orthopyroxenite (Fig. 3b, Electronic Supplementary Materials (ESM) Table S1). Clinopyroxene ($Wo_{40–50}En_{40–41}Fs_{14–15}$) in the ultramafic rocks has higher MgO but lower FeO contents than its counterpart ($Wo_{43–44}En_{44–56}Fs_{2–11}$) in gabbronorite (Fig. 4b). Other oxides of clinopyroxene between the ultramafic rocks and gabbronorite have no significant difference. Plagioclase ($An_{57–69}$) in the ultramafic rocks has slightly higher CaO contents than that ($An_{54–56}$) in gabbronorite (ESM Table S2).

Hornblende grains from the ultramafic rocks have $(Na + K)_A > 0.5$, and therefore are pargasitic hornblende and pargasite with $Fe^{3+} < Al^{VI}$ or magnesio hastingsite with $Fe^{3+} > Al^{VI}$. Hornblende grains in gabbronorite are magnesio-hornblende and actinolitic hornblende with $(Na + K)_A < 0.5$ (Hawthorne et al., 2012). One analysis with the Al_2O_3 content of 4.42 wt% from a slightly altered hornblende grain is not considered here. Hornblende (12.64–14.51 wt%) in the ultramafic rocks has significantly higher Al_2O_3 contents than its counterpart (5.67–9.58 wt%) in gabbronorite (Table 1). The thin hornblende margins in gabbronorite have low Al_2O_3 contents (5.67–6.11 wt%). Biotite from the ultramafic rocks has high MgO contents (21.67–21.84 wt%) than its counterpart (15.20–15.31 wt%) in gabbronorite (ESM Table S3, Fig. 5a).

5.2. Whole-rock geochemistry

Major and trace element contents of the studied samples from the Xiarihamu complex are presented in Table 2 and illustrated in Figs. 6–7. Also displayed here include three samples from the ultramafic rocks with loss on ignition (LOI) < 2.00 wt% (Jiang et al., 2015). Most previous whole-rock major and trace element data from the ultramafic rocks containing > 2.00 wt% LOI and all data from gabbronorite possibly having old ages (423–439 Ma) (Jiang et al., 2015; C.S. Li et al., 2015a; Wang et al., 2014) are not discussed in this study.

The ultramafic rocks show wide ranges of SiO_2 (48.98–54.11 wt%), Fe_2O_3 (0.78–1.61 wt%), CaO (1.53–7.19 wt%), and MgO (22.49–32.16 wt%) contents (Fig. 6). They also have higher MgO, FeO, and Fe_2O_3 contents but lower TiO_2, Al_2O_3, CaO and Na_2O contents than the gabbronorite. The gabbronorite has narrow ranges of SiO_2 contents (51.77–52.16 wt%). The ultramafic rocks and gabbronorite have low K_2O contents (0.08–0.30 wt%), hence showing low K tholeiitic affinities.

The ultramafic rocks and gabbronorite exhibit low total rare earth element (ΣREE) contents and are characterized by flat primitive mantle-normalized REE patterns (Fig. 7a). The ultramafic rocks have lower ΣREE contents (9.58–23.7 ppm) with negligible Eu anomalies (Eu/Eu* = 0.84–1.02) than the gabbronorite (ΣREE = 24.6–33.9 ppm) with positive Eu anomalies (Eu/Eu* = 1.58–2.64). The ultramafic rocks also have higher Cr (1761–2812 ppm), Co (66.0–118 ppm), and Ni

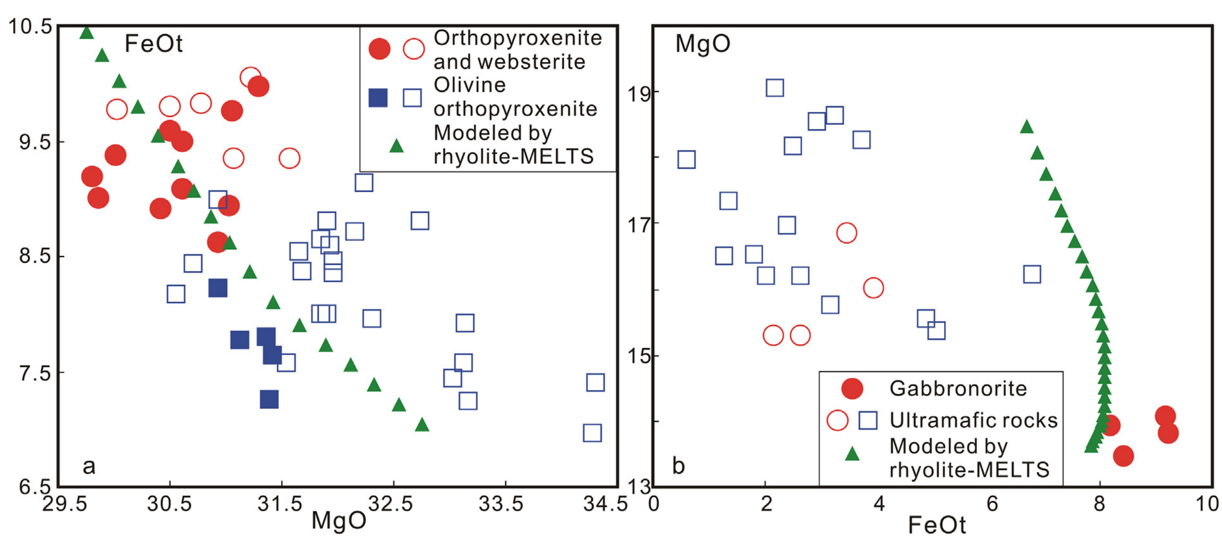

Fig. 4. (a) MgO vs FeOt diagram of orthopyroxene and (b) FeOt vs MgO diagram of clinopyroxene from the Xiarihamu complex and the modeled results by the rhyolite-MELTS program. Solid samples are from this study and hollow samples from Jiang et al. (2015) and C.S. Li et al. (2015a).

Table 1
Chemical compositions of hornblende from selected samples in the Xiarihamu complex.

Sample	X11-2		15X01-5						XRH-b52[a]	
Rock type	Ol orthopyroxenite		Gabbronorite						Ol orthopyroxenite	
SiO_2	45.51	44.38	47.31	47.65	48.14	51.89	51.84	52.82	41.73	43.57
TiO_2	1.61	1.70	1.54	1.49	1.41	0.51	0.52	0.26	2.11	2.06
Al_2O_3	12.64	13.32	9.58	9.32	8.96	5.67	6.11	4.42	14.32	14.51
FeO	4.23	3.73	11.68	11.71	11.58	10.90	10.61	13.94	6.34	5.50
MnO	0.09	0.05	0.15	0.17	0.16	0.16	0.14	0.28	–	–
MgO	17.97	17.64	13.96	14.17	14.10	16.68	16.52	16.94	0.07	0.09
CaO	11.89	12.13	11.47	11.46	11.47	11.27	11.48	8.54	18.81	17.85
Na_2O	2.43	2.38	0.93	0.92	0.89	0.56	0.60	0.43	10.75	11.91
K_2O	0.78	0.86	0.75	0.73	0.64	0.17	0.31	0.08	1.82	2.28
P_2O_5	0.07	0.02	0.00	0.06	0.02	0.02	0.03	0.03	1.22	0.56
Total	97.22	96.20	97.37	97.67	97.37	97.83	98.15	97.76	97.17	98.33
ΔNNO	1.7	1.6	1.1	1.2	1.2	2.3	2.2	2.6	2.2	1.7
T/°C	983	1006	854	847	835	765	771	752	1027	1025
P/kbar	7.1	7.7	4.8	4.5	4.3	1.5	1.8	0.5	8.3	8.4
Classification	Pargasitic hornblende		Magnesio-hornblende			Actinolitic hornblende			Magnesio hastingsite	Pargasite

[a] From C.S. Li et al. (2015a), - Not analyzed.

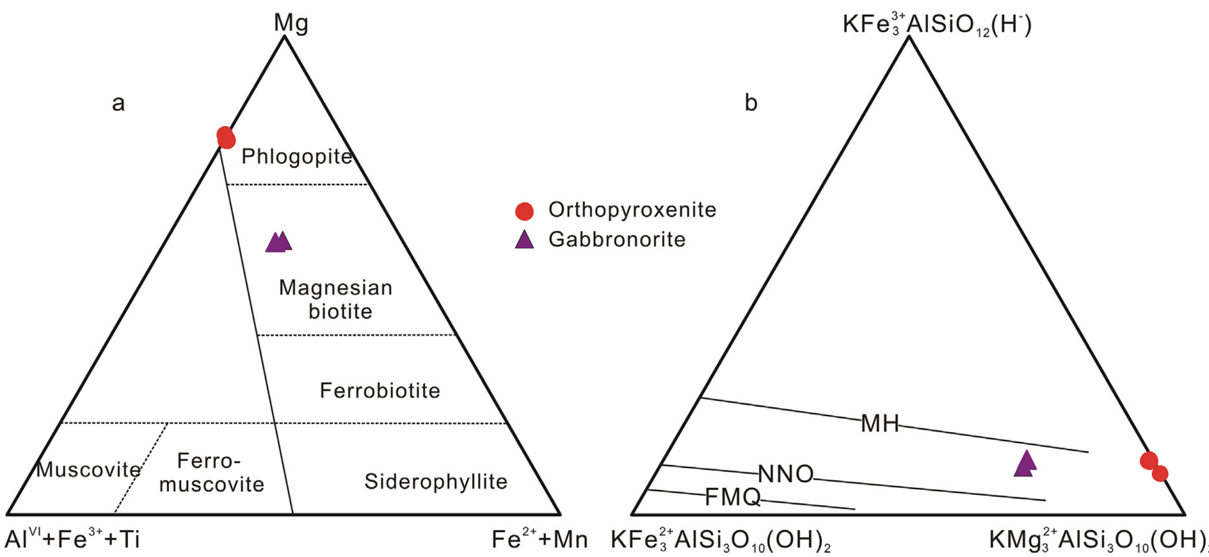

Fig. 5. (a) Biotite classification and (b) oxygen fugacity estimations from biotite (Wones and Eugster, 1965) in the Xiarihamu complex. MH, NNO and FMQ are the magnetite-hematite, nickel-nickel oxide, fayalite-magnetite-quartz oxygen buffers, respectively.

(243–1543 ppm) contents but lower Sr (18.7–84.7 ppm) contents than the gabbronorite (Cr = 33.7–190 ppm, Co = 30.1–34.6 ppm, Ni = 19.5–42.0 ppm, and Sr = 364–465 ppm). All rock types were generally depleted in Nb, Ta, P, and Ti at various degrees in the primitive mantle-normalized spider diagrams (Fig. 7b).

Rubidium-Sr and Sm-Nd isotopic data of this study are presented in Table 3. The Sr-Nd isotopic data of this and previous studies are all showed in Fig. 8. Initial $^{87}Sr/^{86}Sr$ (I_{Sr}) and epsilon Nd ($\varepsilon_{Nd}(t)$) values were calculated at an age of 410 Ma. The ultramafic rocks have wide ranges of I_{Sr} (0.706442–0.710602) and $\varepsilon_{Nd}(t)$ (−2.0 to −7.1). The gabbronorite of 406 Ma shows concentrated I_{Sr} (0.711916–0.712490) and $\varepsilon_{Nd}(t)$ (−3.5 to −3.8) values, but all gabbronorites aged at 406–439 Ma share similar Sr-Nd isotopic characteristics to the 406–411 Ma ultramafic rocks.

6. Discussion

6.1. Pressure, temperature and oxygen fugacity estimation of the Xiarihamu magma

The Na_2O content of magmatic clinopyroxene is pressure-dependent but independent of temperature (Gao et al., 2008). The Na_2O contents of clinopyroxene in the Xiarihamu complex vary from 0.33 to 0.63 wt%, corresponding to crystallization pressures of 5–10 kbar (Nimis and Ulmer, 1998). The hornblende compositions are suitable to calculate the pressure, temperature, and oxygen fugacity of the corresponding subduction-related melts (Ridolfi et al., 2010). Application of the Al-in-hornblende geobarometer of Schmidt (1992) to these compositions yields crystallization pressures of 7.1–8.4 kbar for the ultramafic rocks and 4.3–4.8 kbar for the most hornblende grains from gabbronorite, except for 1.5–1.8 kbar for the hornblende margins from gabbronorite. The temperatures and oxygen fugacities were calculated by using the formulations of Ridolfi et al. (2010) and are 983–1027 °C and ΔNNO +1.6 to +2.2 (NNO is the nickel-nickel oxide oxygen buffer) for the ultramafic rocks, 835–854 °C, ΔNNO +1.1 to +1.2 for the most hornblende grains from gabbronorite, and 765–771 °C and ΔNNO +2.2 to +2.3 for the hornblende margins from gabbronorite. Biotite in the ultramafic rocks has higher Fe^{3+}/Fe^{2+} ratios (determined from biotite stoichiometry) than its counterpart in gabbronorite, corresponding to oxygen fugacities between the NNO and magnetite-hematite (MH) oxygen buffers (Fig. 5b).

Olivine and orthopyroxene show poikilitic texture in the ultramafic rocks and crystallized first and constitute the cumulus that filled or surrounded by minor amounts of biotite, plagioclase, hornblende, and

Table 2
Major and trace element of the selected samples in the Xiarihamu complex.

Sample	15X01-1	15X01-2	15X01-5	15X03-1	15X03-4	15X03-7	15X03-8	15X03-9	1701-2[a]	05-6[a]	05-13[a]
Rock type	Gabbronorite			Websterite	Orthopyroxenite		Ol orthopyroxenite			Websterite	
SiO_2	51.77	51.98	52.16	52.11	54.11	53.46	49.26	50.36	48.98	51.20	51.46
TiO_2	0.44	0.67	0.61	0.31	0.26	0.25	0.22	0.23	0.21	0.42	0.41
Al_2O_3	17.84	17.67	19.60	4.46	4.51	3.87	4.23	4.18	2.92	5.68	6.29
Fe_2O_3	0.71	0.77	0.62	1.38	0.78	1.34	1.49	1.61			
FeO	4.78	5.97	5.47	6.60	7.36	6.99	8.00	7.38			
MnO	0.11	0.13	0.11	0.16	0.16	0.16	0.16	0.15	0.14	0.17	0.15
MgO	6.70	8.27	6.59	23.18	26.77	26.63	30.45	28.56	32.16	23.88	22.49
CaO	12.85	10.25	10.05	7.19	4.01	3.83	2.79	2.89	1.53	6.33	7.11
Na_2O	2.86	2.63	3.32	0.51	0.48	0.38	0.32	0.33	0.84	0.67	0.75
K_2O	0.30	0.26	0.22	0.20	0.14	0.14	0.13	0.08	0.24	0.18	0.21
P_2O_5	0.02	0.04	0.02	0.01	0.01	0.01	0.01	0.01	0.02	0.02	0.03
Cr_2O_3	0.02	0.04	0.01	0.29	0.38	0.38	0.31	0.29			
NiO	0.02	0.01	0.01	0.03	0.05	0.06	0.15	0.10			
SO_3	0.25	0.18	0.07	0.43	0.23	0.15	0.45	0.31			
V_2O_5	0.04	0.03	0.03	0.04	0.03	0.03	0.03	0.03			
LOI	1.37	1.14	1.10	2.30	0.43	2.23	1.78	2.90	1.85	0.84	0.95
Total	100.08	100.04	99.99	99.20	99.71	99.91	99.78	99.41	99.69	99.32	100.32
TFe_2O_3	6.02	7.40	6.70	8.71	8.96	9.11	10.38	9.81	10.80	9.93	10.47
Sc	33.8	25.6	21.9	36.5	29.0	27.1	24.0	26.8	18.1	30.8	31.9
V	216	161	179	206	148	154	138	162			
Cr	92.5	190	33.7	2211	2799	2812	2265	2153	2620	1761	1763
Co	30.1	34.6	31.2	78.4	85.6	82.8	118	103	102	66.0	95.0
Ni	31.8	42.0	19.5	243	385	469	1159	750	1543	337	589
Cu	14.4	12.7	8.03	26.6	46.6	67.0	176	61.6	91.0	50.0	142
Ga	16.1	16.1	17.5	7.05	6.79	6.65	6.14	6.61	5.32	7.61	7.75
Rb	9.84	8.92	6.34	11.6	7.17	6.66	4.95	3.78	16.3	9.86	10.2
Sr	397	364	465	38.6	59.6	38.6	20.4	18.7	24.7	84.7	82.6
Y	14.7	12.1	8.36	7.90	5.32	5.71	4.62	4.78	2.94	6.89	7.21
Zr	22.0	26.2	18.1	20.3	14.9	10.5	13.5	11.6	14.1	26.8	20.9
Nb	0.50	1.97	0.78	0.67	0.52	0.44	0.39	0.34	0.78	0.92	0.74
Cs	1.73	2.06	1.91	2.92	1.47	3.92	1.39	2.14	6.03	2.25	2.66
Ba	71.4	75.0	84.4	34.3	30.8	17.1	19.8	12.1	45.6	52.7	47.5
La	3.26	4.48	3.54	2.28	1.87	1.72	1.27	1.14	2.56	3.24	2.97
Ce	7.83	9.82	7.13	5.00	4.21	4.27	2.74	2.50	5.33	7.68	7.33
Pr	1.36	1.53	1.08	0.73	0.63	0.68	0.42	0.38	0.61	1.05	0.97
Nd	6.96	7.06	4.80	3.51	2.80	3.01	1.74	1.78	2.37	4.45	4.38
Sm	2.22	1.93	1.34	1.14	0.74	0.80	0.50	0.53	0.49	1.14	1.12
Eu	1.23	1.12	1.17	0.35	0.26	0.23	0.18	0.18	0.14	0.42	0.36
Gd	2.55	2.08	1.36	1.24	0.95	0.88	0.61	0.61	0.52	1.42	1.37
Tb	0.44	0.35	0.23	0.22	0.14	0.16	0.11	0.12	0.08	0.29	0.24
Dy	2.76	2.23	1.63	1.42	0.94	0.94	0.75	0.83	0.49	1.48	1.47
Ho	0.55	0.45	0.32	0.30	0.21	0.21	0.18	0.17	0.11	0.37	0.32
Er	1.58	1.30	0.85	0.88	0.60	0.66	0.52	0.53	0.33	0.92	0.92
Tm	0.21	0.19	0.13	0.12	0.11	0.10	0.08	0.08	0.05	0.18	0.13
Yb	1.45	1.22	0.87	0.91	0.72	0.65	0.59	0.62	0.28	0.87	0.84
Lu	0.20	0.18	0.13	0.13	0.10	0.10	0.09	0.10	0.05	0.18	0.13
Hf	0.89	0.91	0.63	0.68	0.50	0.43	0.45	0.43	0.37	0.83	0.67
Ta	0.04	0.14	0.07	0.05	0.03	0.05	0.03	0.03	0.05	0.09	0.05
Pb	9.98	6.19	3.64	4.74	1.46	4.40	3.03	1.78	0.71	3.03	2.43
Th	0.35	0.93	0.52	0.68	0.51	0.41	0.48	0.38	0.77	0.98	0.72
U	0.12	0.29	0.15	0.20	0.13	0.21	0.13	0.10	0.23	0.34	0.21

[a] From Jiang et al. (2015).

pyroxenes. In gabbronorite, rounded or tabular plagioclase grains and absorbed clinopyroxene comprise of the cumulus with minor amounts of interstitial clinopyroxene, plagioclase, hornblende and biotite. The cumulate characteristics are also evidenced geochemically by the high compatible (Cr and Ni) but low incompatible trace element abundances in the ultramafic rocks and by high Sr and Eu but low other trace element abundances in the gabbronorite (Otamendi et al., 2016). Field and experimental studies showed that pyroxenite cumulates with small amounts of olivine generally crystallize from gabbroic melts at high pressure (ca. 10 kbar) (Hermann et al., 2001; Jagoutz and Schmidt, 2013; Villiger et al., 2007). The immediate gneisses that the Xiarihamu complex was emplaced into contain retrograde eclogites aged at 411.1 ± 1.9 Ma (Qi et al., 2014). The crystallization pressures estimated from clinopyroxene (5–10 kbar) and hornblende (4.3–8.4 kbar) in this study are roughly comparable and are also consistent with the crystallization sequence. Considering the fact that hornblende is interstitial and the Al-in-hornblende geobarometer gives the minimum crystallization pressures, the Xiarihamu complex may have experienced a moderate- to high-pressure crystallization. Oxygen fugacity estimates of both this study and C.S. Li et al. (2015a) from Ni concentration in sulfide liquid and the exchange coefficient for Fe and Ni between olivine and sulfide liquid suggest that the complex formed under an oxidized condition.

The gabbronorite from the northern complex and the websterite have a similar zircon U-Pb age (Song et al., 2016), suggesting a genetic relationship of the ultramafic rocks and the gabbronorite investigated. Therefore, the reconstructions of the parental melt and the subsequent magmatic differentiation via combined geochemical and thermodynamical modelings for these cumulates are thus an effective approach (Jakobsen et al., 2005).

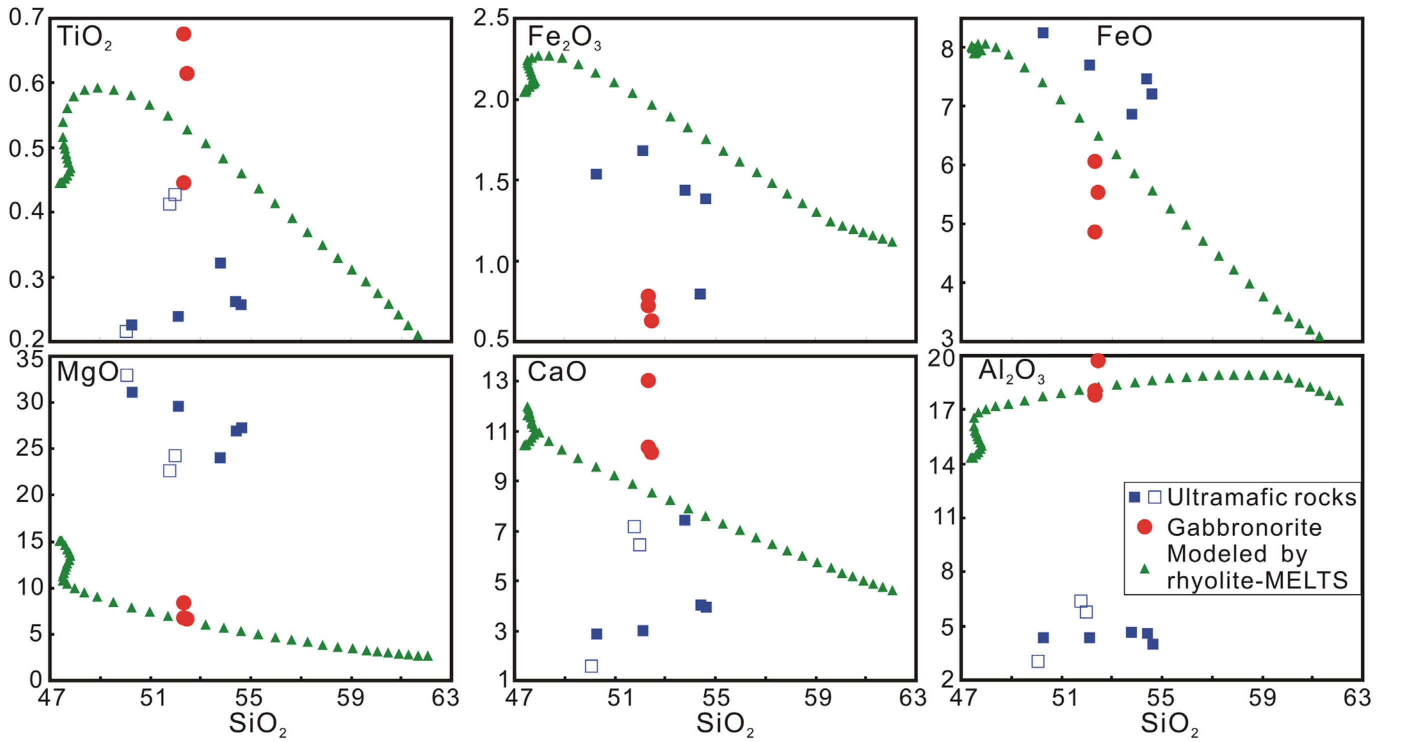

Fig. 6. Major oxide Harker diagrams for rocks from the Xiarihamu complex showing the liquid line of descent modeled by the rhyolite-MELTS program. Solid samples are from this study and hollow samples from Jiang et al. (2015).

6.2. Origin and sources of the parental magma

With the exception of the Sudbury deposits, magmatic sulfide deposits are hosted by a wide range of mantle-derived magmas from komatiitic to flood basaltic, ferropicritic, and high-Al basaltic in composition (Arndt et al., 2005; Naldrett, 2010). Specifically, high-MgO basalt is commonly responsible for magmatic sulfide deposits in post-collisional settings (Gao et al., 2013; Wei et al., 2015; Zhao et al., 2016). As for the Xiarihamu deposit, the characteristics of the parental magma remain controversial and have been proposed to be either high-MgO basalt or boninite (Du et al., 2014; Jiang et al., 2015; C.S. Li et al., 2015a; Song et al., 2016).

An equilibrium distribution method has been developed to calculate the composition of trace elements in the liquid from which cumulates formed (Bedard et al., 2009). This method uses a mass balance equation that reconstructs the composition of the melt from which the cumulus minerals were derived, at the point where this assemblage of cumulates + trapped melt was sealed off from the main body of magma. The method was applied in this study to calculate the Ni contents of the parental melt from which the olivine orthopyroxenite and orthopyroxenite crystallized. The trapped melt fraction is estimated by combined observed results from thin sections and the method of Bedard et al. (2009). The calculated Ni contents of the parental melt are 218–398 ppm with an average of 321 ppm (ESM Table S4), which are similar to the mantle-derived primary magma (Wendlandt et al., 2006) and indicate that an unusually Ni rich parental magma proposed by Song et al. (2016) is unnecessary. Lanthanum, Sm, Dy, and Yb contents of the parental melt were also calculated in order to further constrain the degrees of partial melting from the mantle peridotite because small melting degrees in the mantle will enrich the incompatible elements but deplete the compatible elements relative to large melting degrees (Zhang et al., 2015). The calculated results (exhibited in ESM Table S4 and Fig. 9) suggest that Sm/Yb, La/Yb, and Dy/Yb ratios are low and plot along the melting curve of spinel lherzolite at small to moderate degrees (5–15%), using equilibrium partial melting calculations from the primitive mantle. These results were then used to guide the mantle

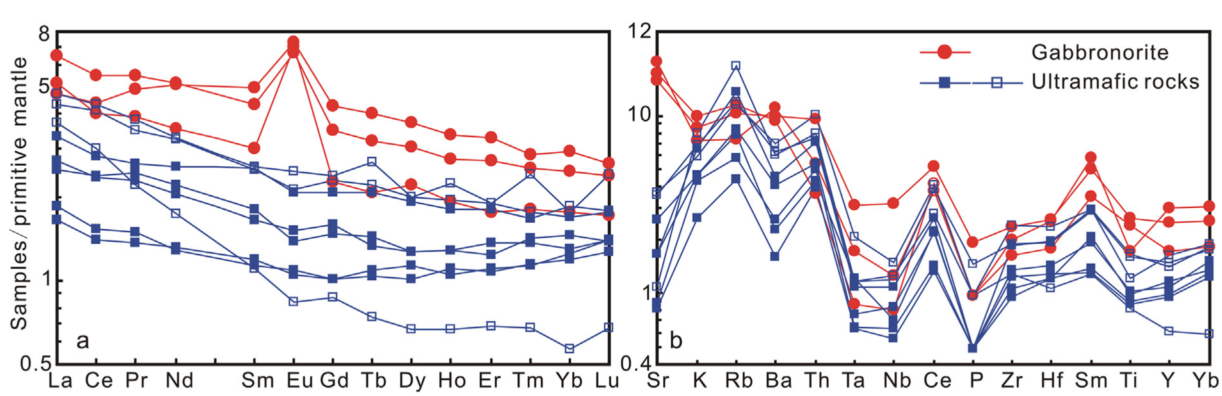

Fig. 7. Primitive mantle normalized rare earth element patterns (a) and trace element spider diagrams (b) for rocks from the Xiarihamu complex. Normalization data are from Sun and Mcdonough (1989). Solid samples are from this study and hollow samples from Jiang et al. (2015).

Table 3
Sr-Nd isotopic compositions of the selected samples in the Xiarihamu complex.

Sample	Rock type	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$	± 2σ	I_{Sr}	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$	± 2σ	$\varepsilon_{Nd}(t)$
15X01-1	Gabbronorite	0.0717	0.712908	6	0.712490	0.1928	0.512449	6	−3.5
15X01-2	Gabbronorite	0.0710	0.712768	5	0.712353	0.1655	0.512360	6	−3.8
15X01-5	Gabbronorite	0.0395	0.712147	6	0.711916	0.1688	0.512385	7	−3.5
15X03-1	Websterite	0.8722	0.713106	12	0.708014	0.1960	0.512453	10	−3.6
15X03-4	Orthopyroxenite	0.3481	0.712150	6	0.710117	0.1595	0.512374	11	−3.2
15X03-7	Orthopyroxenite					0.1607	0.512445	14	−1.9
15X03-8	Ol orthopyroxenite					0.1737	0.512326	9	−4.9
15X03-9	Ol orthopyroxenite					0.1808	0.512441	18	−3.0

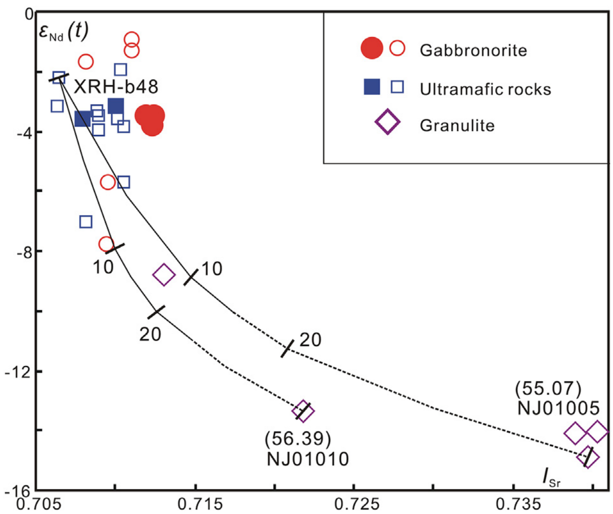

Fig. 8. Initial $^{87}Sr/^{86}Sr$ (I_{Sr}) vs epsilon Nd [$\varepsilon_{Nd}(t)$] diagram showing modeled results. Numbers along the lines and in the brackets represent the degrees of crustal assimilation and the SiO_2 contents of the granulites, respectively. I_{Sr} and $\varepsilon_{Nd}(t)$ values of the granulites are from Yu (2005). Solid samples are from this study and hollow samples from Jiang et al. (2015) and Zhang et al. (2016). Noted that Sr-Nd data of previous studies in the diagram included the gabbronorite in the Xiarihamu complex with ages from 439.1 to 405.5 Ma.

melting model investigated by using the pMELTS program (Ghiorso et al., 2002).

In order to obtain the major oxide compositions of the parental magma, our partial melting modeling using the pMELTS program (Ghiorso et al., 2002) was based on a metasomatized mantle source represented by a spinel lherzolite xenolith (Table 4) (Condamine et al., 2016). Nickel and sulfur contents of the spinel lherzolite were taken as 2000 ppm and 250 ppm, respectively, according to their primitive mantle abundances. The mantle wedge metasomatized by subduction fluids is generally oxidized (Richards, 2015) and thus the oxygen fugacity was taken as ΔQFM +1 (FMQ is the fayalite-magnetite-quartz oxygen buffer). Each isobaric equilibrium partial melting modeling was performed separately at the assumed water contents of 0, 0.1, 0.5 and 1.0 wt%, and pressures of 10, 12, 14, 16, 18, and 20 kbar.

Our modeling results clearly show that H_2O lowers the solidus temperature of the spinel lherzolite and drives the liquids to increase SiO_2, Al_2O_3 and Ni contents and ACNK ratios but to decrease MgO and FeOt contents with almost constant $Mg^{\#}$ ratios. High pressure makes the liquids to contain high MgO, FeOt, and Ni contents and to become high-MgO basalts. The liquids are high-MgO-Al_2O_3 basaltic andesite or andesite under low pressure and hydrous conditions, resembling boninite in major oxide compositions. However, the calculated boninitic liquids contain Ni contents significantly lower than the calculated Ni contents of the parental melt in this study and thus are unaccountable. Song et al. (2016) also suggested that the syn- or post-collisional setting, high-Al_2O_3 spinel, and low-PGE contents of the Xiarihamu complex are unfavorable for the boninitic parental magma. To simultaneously match the Ni contents (~321 ppm) of the parental magma, the melting degrees (~10%) required to form the parental magma, and the melting condition in the spinel stability field, a high-MgO basaltic melt equilibrated with mineral assemblages of Ol (66%) + Opx (25%) + Cpx (7%) + Spl (2%) was calculated at 18 kbar with 0.5 wt% H_2O (Table 4).

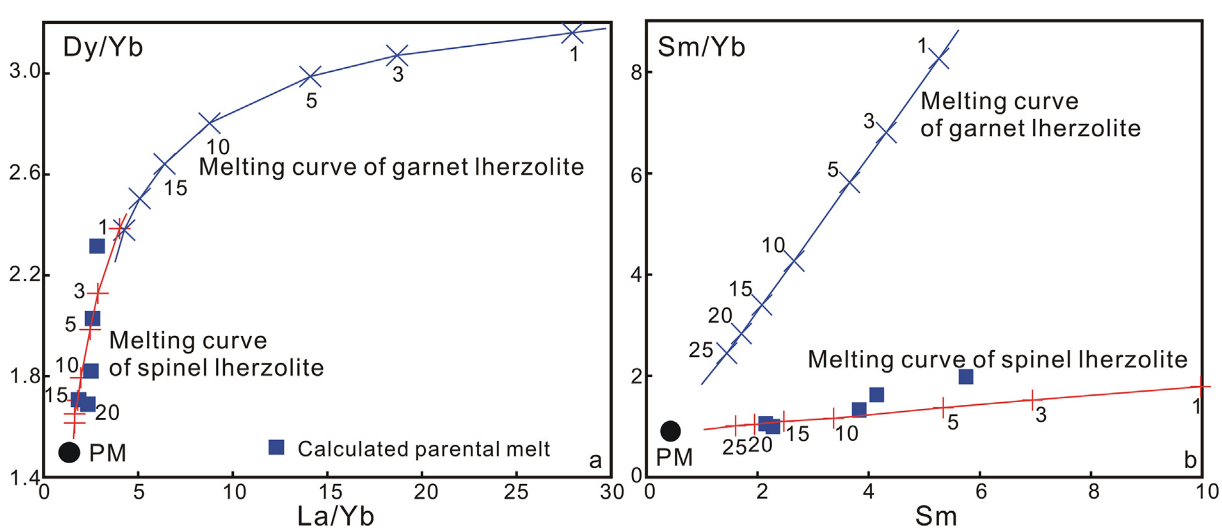

Fig. 9. (a) La/Yb vs Dy/Yb and (b) Sm vs Sm/Yb diagrams for the Xiarihamu parental melt. Melting curves are calculated from equilibrium partial melting modeling of spinel (Ol66 + Opx25 + Cpx7 + Spl2) and garnet lherzolite (Ol60 + Opx20 + Cpx10 + Gt10) from the primitive mantle (PM; Sun and Mcdonough, 1989). Numbers along the lines denote the degrees of partial melting. The partitioning coefficients are taken from the GERM database and Gan (1993). See the text for the calculation about La, Sm, Dy and Yb contents of the parental melt.

Table 4
Major elements of the spinel lherzolite xenolith from Condamine et al. (2016) and of the Xiarihamu parental magma (PM) calculated by the pMELTS program.

No.	Rock type	SiO_2	TiO_2	Al_2O_3	Cr_2O_3	Fe_2O_3	FeO	MnO	MgO	NiO	CaO	Na_2O	K_2O
Br5	Spinel lherzolite	45.01	0.11	3.15	0.34		7.81	0.13	40.57		2.68	0.21	
PM	High-MgO basalt	47.40	0.44	14.34	0.09	1.98	8.09	0.06	15.08	0.04	10.44	1.98	0.05

Therefore, a metasomatized mantle peridotite source with moderate melting degrees in the spinel stability field could produce the high-MgO basaltic melt contained enough Ni contents as the Xiarihamu parental magma.

Song et al. (2016) proposed a pyroxenite mantle source with small melting degrees to produce the Xiarihamu parental high-MgO basaltic magma based on the unusually high Ni/Cu ratios but extremely low PGE contents of the sulfides. Copper is lower than PGE but higher than Ni in partition coefficients between silicates and sulfides (Arndt et al., 2005). Nickel/Cu ratios and PGE contents of the magmatic sulfides depend mainly on both mantle melting degrees and sulfide segregation from silicate melt. Low melting degrees (as low as 6%) would be expected to eliminate mantle sulfides in the oxidized and hydrous mantle wedge metasomatized by subduction fluids (Jugo, 2009) and to generate a melt with low Ni/Cu ratios and high PGE contents. Further melting of this metasomatized mantle peridotite (i.e., moderate melting degrees) will dilute Cu and PGE but increase the Ni/Cu ratios in the melt. A very small degree of olivine crystallization (< 3 wt%) accompanied by limited sulfide segregation during the early magmatic evolution was supported by modeling and petrographical evidence at Xiarihamu (C.S. Li et al., 2015a), and could further increase the Ni/Cu ratios and decrease PGE contents in the melt (Du et al., 2014; Song et al., 2016). Therefore, moderate melting of the metasomatized mantle peridotite combined with limited sulfide segregation during the early magmatic evolution could explain the high Ni/Cu ratios and low PGE contents of the sulfides at Xiarihamu. Olivine from the Xiarihamu ultramafic rocks with the highest forsterite (Fo) contents does not contain the highest Ni contents and negative Fo-Ni correlation for olivine crystals was observed (C.S. Li et al., 2015a). These Fo-Ni relationships clearly suggest Fe-Ni exchange reactions between olivine crystals and the sulfide liquid (Li et al., 2007).

6.3. Evolution of the parental magma

The high-MgO basaltic parental magma deduced from the cumulates in the above section was used to further model the crystallization processes. The rhyolite-MELTS program was used to model isobaric equilibrium crystallization (Ghiorso and Gualda, 2015; Gualda et al., 2012). The nickel and sulfur contents of the high-MgO basalt were taken as 300 ppm and 1000 ppm, respectively. The oxygen fugacity was assumed to be ΔQFM $+1$. The temperature interval during each calculation was between the liquidus temperature computed by the rhyolite-MELTS program and 1000 °C based on the maximum hornblende crystallization temperature.

The modeled results show that orthopyroxene is expected to crystallize after spinel during the initial magmatic evolution at pressures > 9 kbar and that garnet occurs at the end of the magmatic evolution at pressures > 8 kbar. Olivine crystallization persists for the whole magmatic evolution, and the crystallization phase after olivine is clinopyroxene at pressures < 5 kbar. The crystallization sequence modeled along a $P/T = 5$ bar/°C gradient in the pressure interval from 8 kbar to 5.85 kbar is the best fit for that observed at Xiarihamu (ESM Table S5). The pressure interval in the modeling is also consistent with that estimated from the clinopyroxene and hornblende geobarometers. Modeled olivine has the Fo contents of 85–90, completely overlapping the observed Fo values of olivines at Xiarihamu (C.S. Li et al., 2015a). Modeled spinel (crystallized earlier than olivine), orthopyroxene, clinopyroxene, and plagioclase also have the major oxide compositions within the observed ranges for the corresponding minerals in the Xiarihamu complex (Fig. 4).

The evolutionary trend of the modeled melt is strictly controlled by the crystallized minerals (Fig. 6). Small amounts of early crystallized spinel and olivine increase the SiO_2, Fe_2O_3, TiO_2, CaO and Al_2O_3 contents of the residual melt but decrease the MgO and FeO contents. Subsequent crystallization of orthopyroxene makes the balanced melt to have reduced SiO_2, MgO and FeO contents but elevated Fe_2O_3, TiO_2, CaO, and Al_2O_3 contents. Late crystallization of clinopyroxene (plus orthopyroxene) turns the SiO_2, Fe_2O_3, TiO_2 and CaO trends to reverse, and keeps the decreasing MgO and FeO trends and the increasing Al_2O_3 trend (Fig. 6). Spinel is rich in Al_2O_3 contents (as high as 52.47 wt%) and thus suppresses the crystallization of plagioclase. The last appearance of plagioclase obviously decreases the Al_2O_3 contents of the residual melt (Fig. 6).

The Xiarihamu complex contains gneiss xenoliths from the adjacent Jinshuikou Complex. The rocks from the Xiarihamu complex also have highly variable $\varepsilon_{Nd}(t)$ values (-2.0 to -7.1), which can not be explained by an enriched mantle source alone (Hofmann, 1997) but requires crustal contamination. An olivine orthopyroxenite sample with an $\varepsilon_{Nd}(t)$ value of -2.2 and various Early Paleozoic granulites (Yu, 2005; Zhang et al., 2003) from the Jinshuikou Complex have been selected to model crustal assimilation by using the isotopic mixing method of Ma et al. (2000). Calculations show that an amount of 7–9% intermediate granulites is required to explain the variable Sr-Nd isotopic data of the rocks from the Xiarihamu complex (Fig. 8).

The thin margins along the hornblende grains in gabbronorite also have a magmatic origin but are likely to have crystallized from the final interstitial melt. Based on the Al_2O_3 contents (5.67–6.11 wt%) of the hornblende margins, the final interstitial melt is estimated to have a composition similar to quartz diorite and granodiorite (Schmidt, 1992). The estimated composition of the final interstitial melt is comparable with the residual melt compositions obtained by the rhyolite-MELTS modeling (Fig. 6).

Sulfur isotope compositions of the Xiarihamu sulfides are indistinguishable from those of arc basalts originated from the mantle wedge metasomatized by subduction fluids (Alt et al., 1993). Also, the Jinshuikou Complex is composed of gneiss, schist and marble with minor amounts of eclogite, which are all sulfide poor. These lines of evidence supported the notion that crustal sulfur addition for sulfide saturation was unlikely for the Xiarihamu complex. In contrast, the modeled Fe_2O_3 contents of the residual melt shift from increase to decrease. Oxygen fugacities calculated from hornblende and biotite compositions decrease from the ultramafic rocks to gabbronorite. Therefore, we suggest that the decrease of oxygen fugacities induced by crystallization (and perhaps assimilation) was mostly likely responsible for sulfide saturation during the late magmatic evolution. Admittedly, triggers for sulfide saturation in post-collisional magmatic sulfide deposits, such as in the Xiarihamu deposit, remain a subject of debate.

7. Conclusions

Petrographic and mineral compositional data collectively suggest that the rocks in the Xiarihamu complex are cumulates crystallized under oxidized, moderate- to high-pressure conditions. Combined geochemical and thermodynamical modelings suggest that the Xiarihamu parental magma was high-MgO basaltic in composition, generated from a metasomatized mantle peridotite source with moderate melting degrees in the spinel stability field. The high-MgO basaltic parental melt underwent a crystallization process with an early iron enrichment and a late silica enrichment to form the observed crystallization sequence in the complex. Sulfide saturation by crustal sulfur addition is unlikely but was probably caused by magmatic differentiation that induced an oxygen fugacity decrease.

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Appendix A. Supplementary data

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